MATH11219

LECTURE NOTES

Authored by
Lachlan J. W. Wyer

Document Start Date: 27th of Jan 2026
Latest Update: January 29, 2026

Contents

1 Differentiation **2**
1.1 Average Rate-of-Change (ARoC): 2
1.2 Instaneous Rate-of-Change (IRoC): 2
1.3 Limits in Functions: 2
1.4 Common Derivatives: 4

2 Techniques of Differentiation **5**
2.1 Product Rule: 5
2.2 Quotient Rule: 5
2.3 Chain Rule: 5
2.4 Implicit Differentiation 6
2.5 Parametric Differentiation 6
2.6 Logarithmic Differentiation 7
2.7 Higher Derivatives 7

1 Differentiation

Differentiation (Derivatives) is just a mathematical expression of a functions rate-of-change. Think of the gradient of a function, area/estimation of a function, kinematics & optimisation of area or numbers.

1.1 Average Rate-of-Change (ARoC):

The ARoC of the function is just the estimation of a curve based on a linear lines gradient. I.E. the average gradient between two points.

$$\text{ARoC} = \frac{y_2 - y_1}{x_2 - x_1}$$

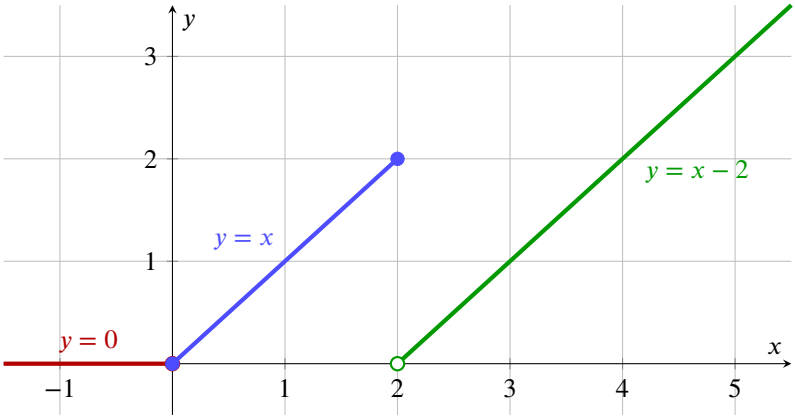
1.2 Instaneous Rate-of-Change (IRoC):

The ARoC of a function is good enough to understand the general direction in which a graph is continuing. Its accuracy can be improved by limiting the amount of change between x_1 & x_2. However a problem occurs when $\Delta x \to 0$. As cannot divide by 0. To find a functions IRoC, use the derivatives by first principles.

$$\begin{aligned}
f(x) &= 3x^2 + 2 \\
f'(x) &= \lim_{\delta x \to 0} \frac{f(x + \delta x) - f(x)}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3(x + \delta x)^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{[3x^2 + 6x\delta x + 3\delta x^2 + 2] - [3x^2 + 2]}{\delta x} \\
&= \lim_{\delta x \to 0} \frac{6x\delta x + 3\delta x^2}{\delta x} \\
&= \lim_{\delta x \to 0} 6x + 3\delta x \\
&= 6x
\end{aligned}$$

1.3 Limits in Functions:

Specifically in Piece-wise functions there can be an issue with derivatives and their interpretations. This is mostly around when the Left and Right Hand Limits not meeting on the same point, making the functions discontinuous.



As can be seem in the graph, at the point $x = 2$ there is not derivative as the function is discontinuous. However there moving in, up until that point a derivative is present. A similar story is true at the point $x = 0$. All other points along the x-axis have a derivative that exists.

1.4 Common Derivatives:

$y = f(x)$	$\frac{dy}{dx}$
c	0
x^n	$n \cdot x^{n-1}$
e^{kx}	$k \cdot e^{kx}$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{\ln(a) \cdot x}$
$\ln(x)$	$\frac{1}{x}$
$\sin(ax+b)$	$a\cos(ax+b)$
$\cos(ax+b)$	$-a\sin(ax+b)$
$\tan(ax+b)$	$a\sec^2(ax+b)$
$\csc(ax+b)$	$-a\csc(ax+b)\cot(ax+b)$
$\sec(ax+b)$	$a\sec(ax+b)\tan(ax+b)$
$\cot(ax+b)$	$-a\csc^2(ax+b)$
$\arcsin(ax+b)$	$\frac{a}{\sqrt{1-(ax+b)^2}}$
$\arccos(ax+b)$	$\frac{-a}{\sqrt{1-(ax+b)^2}}$
$\arctan(ax+b)$	$\frac{a}{1+(ax+b)^2}$
$\sinh(ax+b)$	$a\cosh(ax+b)$
$\cosh(ax+b)$	$a\sinh(ax+b)$
$\tanh(ax+b)$	$a\text{sech}^2(ax+b)$
$\text{csch}(ax+b)$	$-a\text{csch}(ax+b)\coth(ax+b)$
$\text{sech}(ax+b)$	$-a\text{sech}(ax+b)\tanh(ax+b)$
$\coth(ax+b)$	$-a\text{csch}^2(ax+b)$
$\text{arcsinh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2+1}}$
$\text{arccosh}(ax+b)$	$\frac{a}{\sqrt{(ax+b)^2-1}}$
$\text{arctanh}(ax+b)$	$\frac{a}{1-(ax+b)^2}$

2 Techniques of Differentiation

2.1 Product Rule:

The product rule is used to differentiate two functions multiplied together.

$$\begin{aligned} y &= f(x)g(x) \\ \frac{dy}{dx} &= f'(x)g(x) + f(x)g'(x) \end{aligned}$$

$$\begin{aligned} y &= uv \\ \frac{dy}{dx} &= u'v + v'u \end{aligned}$$

2.2 Quotient Rule:

This rule is used when two functions divide each other. It is important to know that the order of functions matter with this rule. As this subtracts rather then addition.

$$\begin{aligned} y &= \frac{u}{v} \\ \frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \end{aligned}$$

2.3 Chain Rule:

By using differentiation by substitution (A.K.A) chain rule, function within other functions can be derived.

$$\begin{aligned} y &= f(g(x)) \\ \frac{dy}{dx} &= g'(x)f'(g(x)) \end{aligned}$$

2.4 Implicit Differentiation

This is used when the equation is not simply $y = f(x)$, e.g.:

$$x^2 - y^3 + \sin(x) - \cos(y) = 1$$

$$\begin{aligned}
\frac{d}{dx}(x^2) &= 2x \\
\frac{d}{dx}(-y^3) &= -3y^2\frac{dy}{dx} \\
\frac{d}{dx}(\sin(x)) &= \cos(x) \\
\frac{d}{dx}(-\cos(y)) &= \sin(y)\frac{dy}{dx} \\
\frac{d}{dx}(1) &= 0
\end{aligned}$$

$$\begin{aligned}
\frac{d}{dx}(x^2 - y^3 + \sin(x) - \cos(y)) &= \frac{d}{dx}1 \\
2x - 3y^2\frac{dy}{dx} + \cos(x) + \sin(y)\frac{dy}{dx} &= 0 \\
\frac{dy}{dx} &= \frac{-2x - \cos(x)}{\sin(y) - 3y^2} \\
&= \frac{2x + \cos(x)}{3y^2 - \sin(y)}
\end{aligned}$$

2.5 Parametric Differentiation

Basically just multiplying the derivatives together.

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

2.6 Logarithmic Differentiation

Just follow the basic steps and the derivative will be found. Mostly useful for a short version of the product rule.

1. Find the log of each side of the equation, & use log laws to expand the equation
2. Differentiate each side with respect too the independent variable
3. Rearrange and substitute
4. Simplify

2.7 Higher Derivatives

Found by differentiating the function multiple times.
These are useful as they can describe the nature of a function, such as
If $\frac{dy}{dx} > 0$, the gradient can be said to be increasing, or $\frac{dy}{dx} < 0$, the gradient is decreasing and $\frac{dy}{dx} = 0$ the gradient is stable.
Or $\frac{d^2y}{dx^2} > 0$, concavity is up and the point (x,y) is described as a minimum. $\frac{d^2y}{dx^2} = 0$, described as a Point of Inflection. Basically just the point where the concavity of a function changes. $\frac{d^2y}{dx^2} < 0$, concave down and point (x,y) is a maximum.